

Abstract

This study translates DeepSolo's ordered-points, text-boundary, and reading-order representation into a Korean OCR correction strategy on top of Apple Vision OCR. The official DeepSolo inference pipeline could not be fully reproduced on macOS arm64 because of CUDA and Detectron2 dependencies. Instead, this work reproduced the output representation as a proxy: each Apple Vision observation is converted into center-line ordered points, boundary points, and ordered text units. The resulting pipeline applies ROI cropping, rotation, contrast/upscale preprocessing, ordered-point-based reading-order correction, expected-vocabulary reconstruction, and fuzzy correction to raw Apple Vision OCR outputs. For decorative vertical titles that Vision fails to detect at all, the system adds a pre-OCR ordered spot proposal branch that crops and normalizes the title region before recognition.

The core hard Korean set was expanded from four real failure seeds to N=20 by adding sixteen controlled vertical sign/poster cases. On this set, raw CER was 0.839, while the pre+post pipeline reduced CER to 0.006. The full experiment was re-aggregated under four conditions: raw Vision, preprocessing only, postprocessing only, and preprocessing plus postprocessing. In the title-critical Canva/MiriCanvas subset (N=24), raw CER was 5.250, post-only CER was 5.250, and pre+post CER was 0.080, showing that missing decorative titles require an ordered spot proposal before OCR. The single public-safe "Gwangbokjeol" missing-title probe is included in the CER aggregate and is also reported by target recall. Its raw/post-only recall was 0.000/0.000, while pre-only/pre+post recall was 1.000/1.000. Over the Main GT set (N=784), raw CER was 1.209, pre-only CER was 0.067, post-only CER was 1.096, and pre+post CER was 0.018.

Keywords: DeepSolo, text spotting, Apple Vision OCR, Korean OCR, vertical text, reading order, HangulGuidedOCR

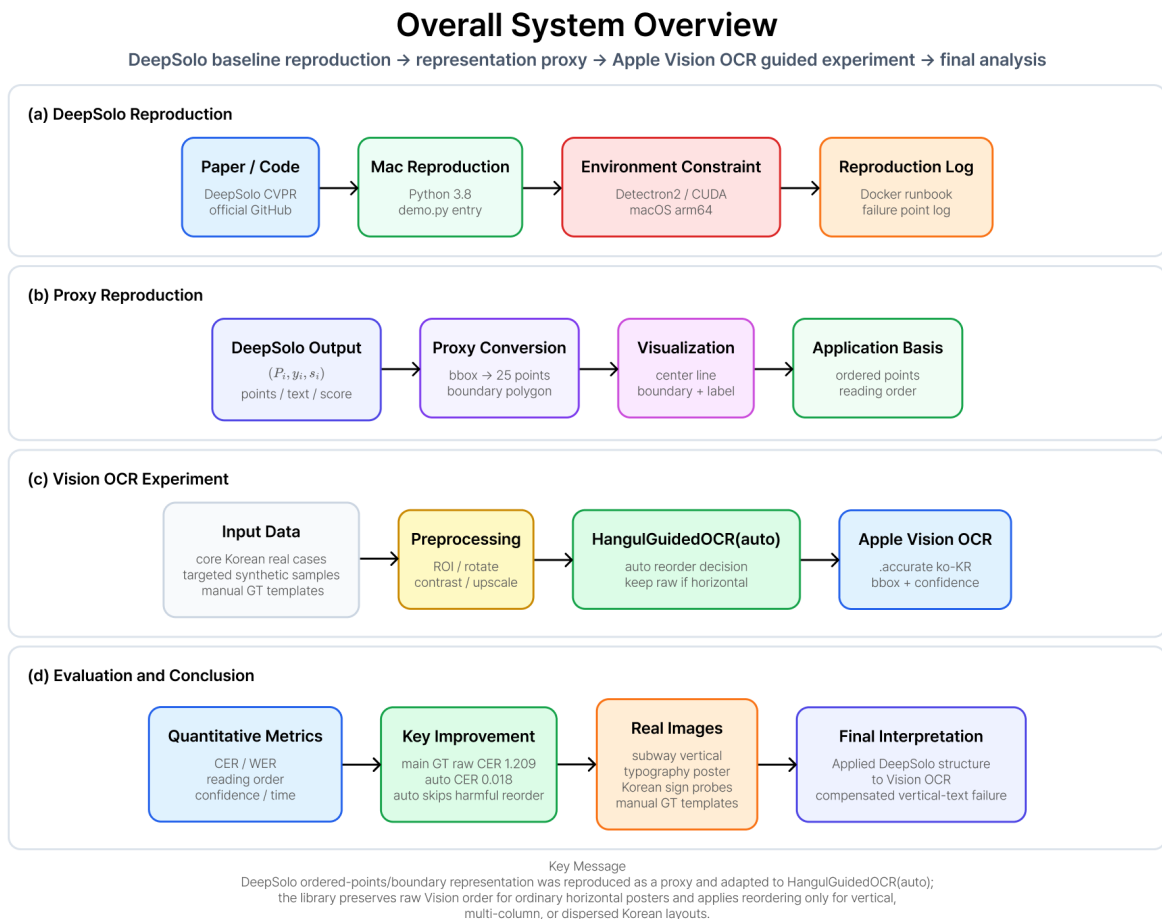


Figure 1. Overall research flow: DeepSolo reproduction, proxy representation, Vision OCR experiment, evaluation, and conclusion.

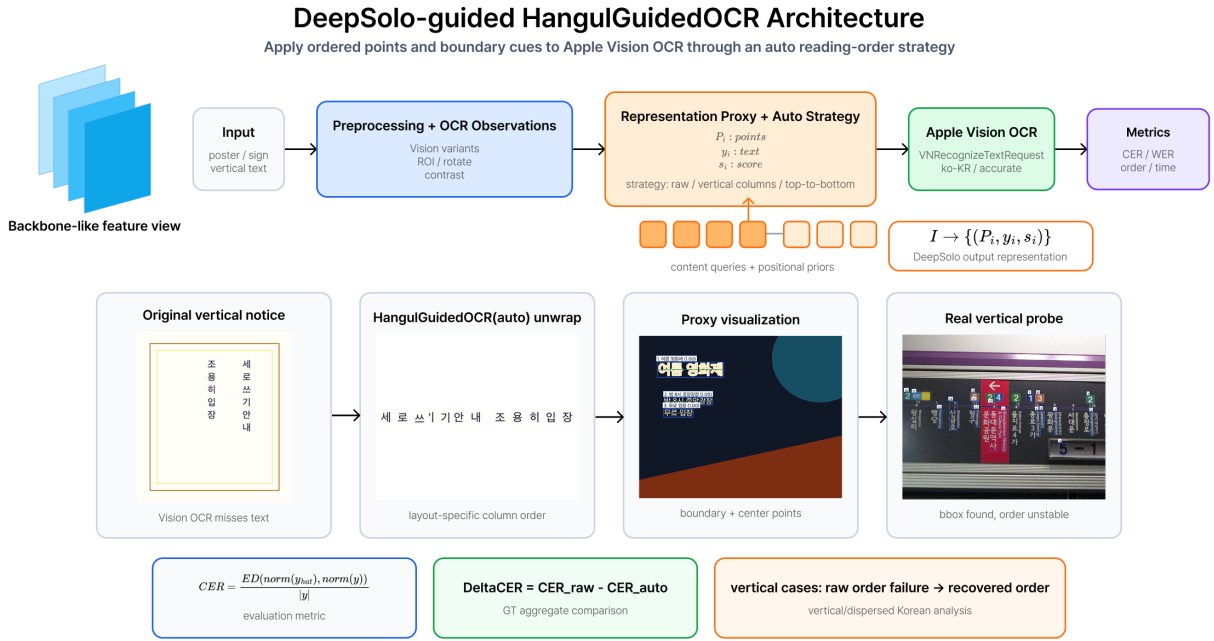


Figure 2. System overview that converts the DeepSolo representation into an Apple Vision OCR correction structure.

1. Introduction

OCR is a core mobile-app function that converts camera input into text, but Korean posters, signs, and notices are much more complex than ordinary document OCR. For vertically written station names such as subway signboards, direction and semantic reading order are critical, and poster-style dispersed typography does not follow ordinary line structure. Apple Vision OCR is strong on controlled horizontal text, but in these layouts, even when observations are detected, they may not be assembled into the string order expected by a human reader.

The goal of this project is not to port the DeepSolo model itself to macOS, but to transfer DeepSolo's text-instance representation, namely ordered points and text boundaries, into Apple Vision OCR input correction and output postprocessing. Therefore, the experiment focuses less on dataset size alone and more on interpreting where Korean OCR actually fails and which layout priors are required.

The study asks three questions. First, how far can the official DeepSolo baseline be reproduced in the current Mac environment? Second, if the DeepSolo representation is reproduced as a proxy, can it explain and correct Korean reading-order failures in Apple Vision OCR? Third, can this correction strategy be generalized into a Swift/Xcode library API usable by app developers?

2. Related Work

Scene text spotting research has developed from RoI-based architectures that separate detection and recognition, to segmentation-based architectures, and then to Transformer-based end-to-end architectures. CRNN is a representative CNN-RNN-CTC scene text recognizer, and ASTER is often compared as a recognition baseline combined with rectification. TextBoxes, EAST, CRAFT, and DBNet stabilized scene text detection, while ABCNet and TESTR addressed curved text and end-to-end spotting. Within this line of work, DeepSolo represents each text instance with explicit points and a boundary-aware representation, lifting center-line, boundary, and order information that

simple rectangular boxes tend to lose into the model's internal representation.

From the OCR-engine perspective, systems such as Apple Vision OCR, Tesseract, EasyOCR, and NAVER CLOVA OCR are strong on document-like and horizontal text, but they do not always guarantee the reading order expected by people in Korean posters and signs that contain vertical writing, multiple columns, or dispersed typography. Existing OCR APIs return recognized strings and position observations, but there is relatively little Swift library support for systematically converting those outputs into a DeepSolo-style ordered point representation when Korean app developers already know domain vocabulary and layout priors, such as station boards, notices, or menus.

Reading-order estimation and Korean OCR are also relevant. LayoutReader shows that document reading order can be modeled as a layout-aware ordering problem rather than as a simple left-to-right text concatenation task. Korean OCR work such as Hangul character decomposition addresses the combinatorial structure and class imbalance of Hangul syllables, while KORIE-style Korean receipt benchmarks highlight that Korean OCR evaluation must consider domain-specific layout and transcription rules. These studies motivate this report's focus on Korean layout priors: subway station labels, vertical notices, poster titles, and dispersed typography require different ordering assumptions even when the underlying OCR engine is unchanged.

Instead of training a new OCR recognizer, this study converts DeepSolo's ordered-points/boundary perspective into an Apple Vision OCR preprocessing and postprocessing structure. Thus, the contribution is not the direct reproduction of DeepSolo model performance, but the translation of a text spotting representation into Korean reading-order correction and the design of the HangulGuidedOCR API for Xcode.

3. DeepSolo Reproduction and Proxy

DeepSolo represents text instances in a Transformer encoder-decoder architecture using explicit point queries and point priors. In the paper, the output for an image I is formulated as $\{(P_i, y_i, s_i)\}$, where P_i can

be interpreted as points connected to the text centerline/boundary, y_i as the recognized string, and s_i as the confidence score. The basic structure creates Top-K Bezier center-curve proposals, uniformly samples $N=25$ ordered points from each curve, and feeds the decoder with a combination of content queries and positional queries. The decoder predicts text/background scores, character classes, center points, and boundary points, and the final string is produced through CTC decoding.

The official GitHub code depends heavily on Detectron2, CUDA, and specific PyTorch/CUDA combinations. Because the experimental environment in this work is macOS arm64, CUDA inference could not be executed directly, and Docker/UTM reproduction did not reach actual model inference because of dependency resolution and the absence of GPU acceleration. Therefore, the reproduction record was divided into official inference reproduction, which preserves failure logs, and representation reproduction, which reproduces the paper's output representation as an implementable proxy.

In the proxy reproduction, position observations returned by Apple Vision were converted into 25 center-line ordered points and a boundary-point polygon. These values are not DeepSolo's learned point queries, so they cannot be claimed as a reproduction of model performance. Instead, they are a structural reproduction that explains why a DeepSolo-style representation is useful for Apple Vision OCR correction. In other words, Vision observations are converted into text-unit position, order, and boundary representations, and those representations are used for reading-order sorting, ROI restriction, duplicate point suppression, and CTC-like repeated-character collapse. When OCR fails to detect text at all, an additional branch specifies an ordered spot proposal before OCR rather than after Apple Vision observations are produced.

official reproduction: demo.py and the Docker runbook were inspected, but reproduction stopped because of CUDA/Detectron2/macOS arm64 constraints.

proxy reproduction: Vision observations were converted into 25 center-line ordered points, boundary points, and label visualizations.

point decoding: repeated point predictions are cleaned with CTC-like collapse, for example "관광광복절 -> 광복절".

application: this proxy representation was connected to ROI, reading order, expected-vocabulary correction, and HangulGuidedOCR API requirements.

4. Method

The overall pipeline consists of input images, Vision OCR variants, a DeepSolo-style representation proxy, Apple Vision OCR results, and metrics. At the input stage, the experiment applied not only full-image OCR but also ROI crop, rotation, contrast adjustment, and upscale2. At the representation-proxy stage, OCR observations were converted into ordered points and boundary points, and text components were reordered according to the layout strategy.

Given an Apple Vision observation $o_i=(B_i, y_i, s_i)$, let the four corners of B_i be b_{tl} , b_{tr} , b_{br} , and b_{bl} . For a horizontal or rotation-corrected text strip, define the two endpoints of the centerline as $c_0=(b_{tl}+b_{bl})/2$ and $c_1=(b_{tr}+b_{br})/2$. Then sample $N=25$ ordered center-line points as $p_k=(1-t_k)c_0+t_k c_1$, where $t_k=k/(N-1)$, $k=0, \dots, N-1$. Here $P_i=\{p_0, \dots, p_{24}\}$, y_i is the top OCR candidate, s_i is the confidence, and the boundary proxy is preserved as $B_i=\{b_{tl}, b_{tr}, b_{br}, b_{bl}\}$. For

vertical columns or rotated ROIs, the same equation is applied in the corrected coordinate system and then mapped back to the original coordinate system.

Algorithm	Procedure
bbox→points	Input observation $o=(B, y, s)$; order B as tl, tr, br, bl ; set $c_0=(tl+bl)/2$ and $c_1=(tr+br)/2$.
sampling	For $k=0..24$, set $t=k/24$ and $p_k=(1-t)c_0+t c_1$; return $P=\{p_k\}$, boundary B , text y , score s .
ordering	Group point proxies by candidate layout; sort by x/y center according to horizontal, vertical $L \rightarrow R/R \rightarrow L$, or top-to-bottom strategy.
decoding	Suppress overlapping proposals, then collapse repeated adjacent labels so 관광광복절 becomes 광복절.

Algorithm 1. Procedure for converting a Vision bbox into a DeepSolo-style ordered-point proxy.

Algorithm	Procedure
proposal	Input image I , ROI/layout prior R , expected terms E , orientation hint θ .
normalize	For each ROI $r \in R$, crop $I_r = \text{crop}(I, r)$, rotate or unwrap by θ , then apply upscale/contrast when needed.
pre-OCR	Run Apple Vision on I_r before full-image ordering; convert detected boxes to P_i with Algorithm 1.
gate	If target recall is low or the main title is missing, keep the proposal; otherwise preserve raw Vision order.
merge	Merge proposal text with raw observations using expectedOrder/fuzzy vocabulary only when domain terms are supplied.

Algorithm 2. Pre-OCR ordered spot proposal generation and auto-gate flow.

Reading strategies are divided into horizontal-lines, vertical-columns-left-to-right, vertical-columns-right-to-left, and top-to-bottom. HangulGuidedOCR is always called as an OCR wrapper, but its internal auto strategy determines the layout. For a vertical subway station sign, after a 90-degree rotation, the output should be assembled as "동대문역사 문화공원", matching the station order implied by the top-to-bottom English annotation. For user_typography_wall, the text should be read from top to bottom as "예쁜 그림을 그리고 싶다". For ordinary horizontal posters, HangulGuidedOCR internally preserves the raw Vision order and skips reordering.

The auto strategy is based on the shape and distribution of ordered point proxies sampled from the observed OCR observations. If there is one or fewer observations, the raw order is preserved. If there are at least two tall point sequences satisfying $\text{height} > 2.0 \times \text{width}$ and the x-direction center spread exceeds 0.10, the sample is treated as a vertical-columns candidate. If row grouping yields at least the expected term count, the y spread exceeds 0.18, and y spread $> 0.75 \times \text{x spread}$, the sample is classified as top-to-bottom rows. Most ordinary posters and card news samples that do not satisfy these conditions remain in rawVisionOrder, preventing unconditional point sorting from damaging paragraph order.

Expected-vocabulary correction uses fuzzy matching when the substring detected by OCR is close to a target vocabulary item. This correction is not intended to hallucinate arbitrary sentences. It is used only in restricted cases where the app developer already knows domain

vocabulary, such as subway station names, menu items, or notice keywords.

Case	Auto action	Expected output
Horizontal poster	Keep raw order	Preserve raw-level CER
Subway station	Rotate + expectedOrder	동대문역사 문화공원
Top-to-bottom	Row sort + vocabulary	예쁜 그림을 그리고 싶다
Known terms	Fuzzy correction	Recover app-domain vocabulary

Table 1. Internal layout-specific decision rules for HangulGuidedOCR(auto).

5. Experimental Setup

The experiment was divided into six layers. First, the Core hard Korean set contains 20 samples: four existing real failure seeds and sixteen new controlled hard expansions in vertical sign/poster form. The new sixteen samples are not independently photographed real photos; they are controlled failure probes that place Korean vertical-writing or sign-like phrases with known ground-truth transcriptions on local poster, sign, or subway backgrounds. Second, the 600 targeted Korean poster synthetic samples control layout variables such as large vertical titles like "광복절", brush-style vertical posters, vertical subway station boards, and dispersed typography. Third, the 124 Canva/MiriCanvas local templates are real design-template-based samples manually transcribed by the user. Fourth, among these local templates, a title-critical subset of 24 samples with severe title omission or distortion was separated to measure large-title failures that are diluted in full-poster CER. Fifth, 15 public Korean sign samples with user-filled GT were included as an auxiliary real-sign evaluation. Sixth, one public-safe "광복절" missing-title probe with no copyright issue was included to separately confirm the need for a pre-OCR ordered spot proposal.

The quantitative metrics are CER (Character Error Rate), WER (Word Error Rate), term recall, and processing time. CER was calculated as $ED(\text{norm}(\hat{y}), \text{norm}(y))/|y|$, and mixed Korean/English/number samples were processed with NFC normalization and whitespace cleanup. For public Korean signs, only rows with user-filled manual GT were included, and duplicate image hashes such as 0012 and 0013 were removed.

CER was not clipped to the range between 0 and 1. In edit-distance-based CER, if the GT is a short title while OCR outputs long surrounding text or numbers together, insertion errors can exceed the number of GT characters, so CER can be greater than 1. Therefore, the large CER values in the title-critical and stylized-title results indicate the severity of simultaneous title omission and surrounding-text over-detection.

Set	N	GT	Role
Core hard	20	Y	4 real seeds + 16 hard expansions
Target synth	600	Y	Controlled vertical titles/stations/multi-column layouts
Manual templates	124	Y	User-transcribed posters
Title-critical	24	Y	Large-title omission local probe
Public signs	15	Y	Manual-GT public Korean signs
Stylized title	1	Y	Public-safe title-omission probe

Table 2. Composition of the main experiment and auxiliary probes.

6. Results

Set	N	Raw	Pre	Post	Full
Hard expansion	16	0.947	0	0.938	0
Real seed	4	0.408	0.0278	0.408	0.0278
Title-critical	24	5.25	1.68	5.25	0.0799
Manual templ.	124	0.0602	0.0602	0.191	0.0602
Public signs	15	0.283	0.283	0.284	0.284
Stylized title	1	14	0.333	14	0
Target synth	600	1.3	0	1.13	0
Main GT weighted	784	1.21	0.0669	1.1	0.0175

Table 3. CER comparison under raw Vision, preprocessing-only, postprocessing-only, and preprocessing-plus-postprocessing conditions. The single missing-title probe is included in the main GT average, and target recall is interpreted separately in Table 4.

The main aggregate should be interpreted together with both mean and standard deviation. Across the full Main GT set, raw CER was 1.209 ± 1.574 , pre-only CER was 0.067 ± 0.374 , post-only CER was 1.096 ± 1.611 , and pre+post CER was 0.018 ± 0.078 . The standard deviation is large because subsets with very different failure intensities, such as ordinary horizontal templates and missing-title posters, are included together. Therefore, the single aggregate CER value should be interpreted as a weighted summary of the overall trend, and actual performance should be judged together with per-subset results.

Case	Raw	Pre	Post	Full
광복절	0.000	1.000	0.000	1.000

Table 4. Target recall for the missing-title probe that requires a pre-OCR ordered spot proposal.

Mode	CER	WER	Decision
Raw Vision	0.060	0.087	baseline
Forced proxy	0.191	0.238	order harmed
HG OCR(auto)	0.060	0.087	keep raw

Table 5. Auto-gate ablation showing why forced proxy reorder is risky for ordinary horizontal posters.

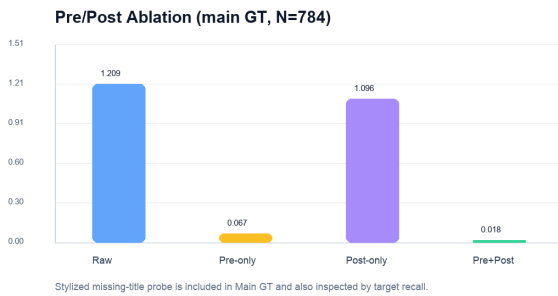


Figure 3. CER comparison across Raw, Pre-only, Post-only, and Pre+Post conditions on the main GT set of 784 samples.

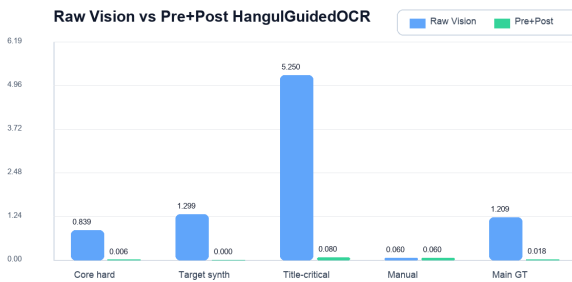


Figure 4. CER change of HangulGuidedOCR(auto) compared with raw Vision on GT-annotated sets.

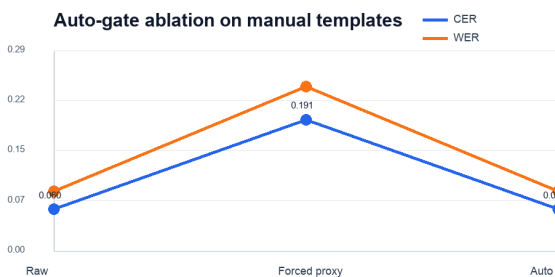


Figure 5. Comparison of forced proxy reorder and auto decision on the Canva/MiriCanvas transcription set.

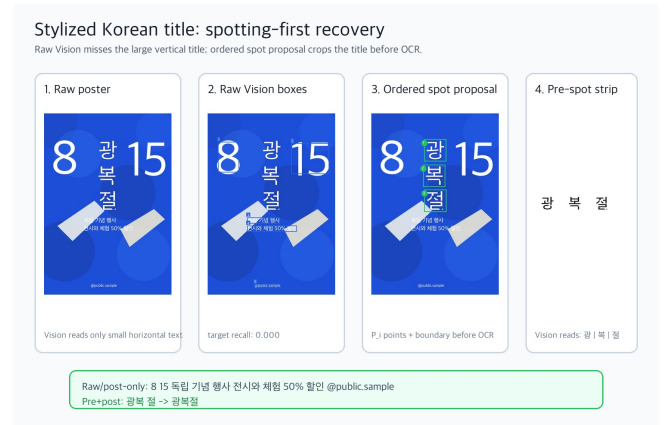


Figure 6. Recovery process for raw OCR omission in a decorative vertical title using a pre-OCR ordered spot proposal.

The Core hard Korean set is an N=20 set combining four existing real failure cases and sixteen controlled hard poster/sign expansions. In the existing real cases, the raw output for the subway station sign remained only at `동대문역사`, or dispersed typography was incorrectly assembled as `형 싶 다 왜`. The final target order for the subway case was corrected to "동대문역사 문화공원", matching the top-to-bottom order of the English annotation. In the sixteen new vertical-writing/sign-like samples, raw Vision mostly failed to detect the vertical phrases, producing an average raw CER of 0.947 ± 0.212 , while the guided/unwrapped input produced by the ordered-point proxy achieved CER 0.000 ± 0.000 . However, these sixteen samples are controlled hard expansions rather than independently photographed real photos, so they should be interpreted as failure-condition augmentation rather than as a real-photo benchmark.

In contrast, the 124 manually transcribed Canva/MiriCanvas samples were mostly ordinary horizontal posters and card news, and Apple Vision raw already performed well. On this set, applying ordered-point proxy reorder unconditionally damaged natural paragraph order and worsened average CER from 0.060 to 0.191. This forced-proxy result is not the final method; it is an ablation that shows why the auto gate is necessary. HangulGuidedOCR(auto) classified these samples as horizontal layouts and skipped reordering, preserving CER at 0.060. Therefore, the final comparison is not "raw Vision vs forced proxy", but "raw Vision vs always-called HangulGuidedOCR(auto)".

For the title-critical subset of 24 Canva/MiriCanvas samples where large titles are omitted or distorted, full-poster CER dilutes the failure because of the length of body text. Therefore, an additional evaluation was added with only the title region as the target. In this subset, raw CER was 5.250 and post-only CER was also 5.250, showing almost no recovery, while pre+post CER decreased to 0.080. Thus, for missing titles such as "광복절", the key factor is an ordered spot/proposal before OCR rather than ordering after OCR.

The targeted Korean poster synthetic set was regenerated with 600 samples, most of which are vertical-title or station-board types such as stylized vertical title, vertical brush poster, vertical station board, and vertical notice poster. On this set, raw CER was 1.299 and pre+post CER was 0.000. In addition, 15 public Korean sign samples with user-filled manual GT were included after duplicate images were removed; because this set did not include separate ROI/proposal annotations, the difference between raw and pre/post was small.

The Main GT weighted value in Table 3 is not the simple average of per-set means, but a sample-count-weighted average. The calculation includes Real seed, Hard expansion, Target synthetic, Manual templates, Title-critical templates, Public Korean signs, and the single public-safe "광복절" missing-title probe. This probe is also reported with target recall in Table 4 to clearly show the nature of the title-detection failure.

Sample	Raw	Auto	Guided text
subway_dongdaemu	0.444	0.000	동대문역사 문화공원
user_typography_	0.900	0.000	예쁜 그림을 그리고 싶다
real_busy_street	0.091	0.091	장국전문 비빔밥 아침식사 생태찌개
user_movie_poster	0.263	0.000	악하게 재미있다 올해 최고의 영화

Table 6. Representative recovery results for core real failure samples.

Subway vertical sign recovery
세로 역명판 ROI를 옆으로 돌린 뒤, 검출된 두 한글 열을 한국어 역명판 읽기 순서로 재배열한다.

(a) Vertical Korean ROI

(b) Rotate 90° + ordered points

(c) Guided reading order

(d) Result

Recovered output
raw: 2 4 & Culture Park 동대문역사 문화공원
auto: 동대문역사 문화공원
target recall: 0.000 → 1.000

Figure 7. Transformation process for the subway vertical station sign.

Typography poster recovery

분산된 한글 component를 위에서 아래로 정렬한 strip으로 만들고 기대 문장 보정으로 복원한다.

(a) Original poster

(b) Ordered points overlay

(c) Component strip + correction

ordered component strip

예쁜 그림을 그리고 싶다

raw strip OCR: 예쁜 그림을 그리고 싶다
auto: 예쁜 그림을 그리고 싶다

(d) Result

Recovered output
raw full image CER: 0.900
component strip CER: 0.100
auto corrected CER: 0.000

Figure 8. Transformation process for the dispersed typography poster.

7. Discussion

The key point of this experiment is that internal automatic judgment is necessary when HangulGuidedOCR is always used. For vertical writing and dispersed typography, even if the OCR engine detects some characters, the string assembly order can be wrong, so ordered point sequences and boundary priors are effective. Conversely, for ordinary horizontal posters, Vision already preserves paragraph order well, so HangulGuidedOCR(auto) should not apply point-proxy reorder.

This result directly informs the design of the HangulGuidedOCR library. The library adds image transforms, reading strategies, and expectedOrder correction on top of a raw Vision wrapper, and in auto mode it determines whether to apply reordering based on layout evidence. If a developer knows a layout prior, such as a subway station sign, vertical notice, or dispersed typography, the strategy can be specified explicitly; even when it is not specified, ordinary horizontal inputs are handled in a way that preserves raw order.

Therefore, the success of this study is not that it applies reordering unconditionally. Its success is that when HangulGuidedOCR is always adopted, the internal auto strategy distinguishes between cases that require improvement and cases where raw order should be preserved. This is crucial for real app development: a library must improve difficult vertical/multi-column samples while not breaking ordinary posters that already work well.

8. Library-oriented System Implementation

The research results are extended into HangulGuidedOCR, a separate SwiftPM repository. Rather than listing usage steps, this section summarizes the design principles derived from the experiments. First, the library does not replace Apple Vision OCR; it wraps it. Second, Vision observations are not used as raw bbox sorting units, but are converted through HangulPointProxyBuilder into ordered point proxies of the form `(P\_i, y\_i, s\_i)`. Third, correction is not forced on every input; in auto mode, reading-order correction is applied only when layout evidence is sufficient. Fourth, when the developer knows domain vocabulary such as subway station names, menu items, or notice keywords, expectedOrder/fuzzy correction can be passed explicitly.

Vision wrapper: runs ko-KR OCR based on VNRecognizeTextRequest and returns text observations and confidence.

Spotting proposal: passes ROI boundary, 25 ordered points, score, and rotation hint through `HangulSpottingProposal` before OCR.

Point proxy: converts each observation into 25 center-line ordered points and 4 boundary points.

Point decoding: reduces overlapping proposals through overlap suppression and cleans repeated point characters with CTC-like collapse.

Image transform: ROI crop, rotation, upscale, and contrast preprocessing.

Reading strategy: horizontal, vertical L->R, vertical R->L, and top-to-bottom.

Correction: expectedOrder, target terms, and fuzzy vocabulary correction.

Tests: subway vertical station board and user typography wall recovery.

The performance overhead depends on the selected preprocessing operations and the number of Vision OCR runs. For ordinary horizontal inputs, raw order is preserved, so the additional cost is limited to observation analysis and string reconstruction. For vertical-writing/sign-like inputs, ROI crop, rotation, and unwrapped strip OCR may take longer than a single raw OCR pass, but as shown in the Core hard set, the benefit of converting a failed phrase into a detectable input is more important. Therefore, the library is designed to allow explicit strategies for domain-specific inputs where accuracy is important, while auto mode preserves raw order for ordinary posters.

The library repository and the research repository are separated. The research repository preserves experimental artifacts, the paper-style report, and DeepSolo reproduction records, while the HangulGuidedOCR repository is maintained as a product-facing package that app developers can import. Both READMEs share the same conclusion, but the former focuses on experimental evidence and the latter on API usage.

9. Limitations and Future Work

First, the official DeepSolo inference pipeline was not fully reproduced in a CUDA Linux environment. Therefore, the DeepSolo result in this report is a representation-proxy reproduction, not a model-performance reproduction. Second, the Core hard set was expanded to 20 samples, but the new sixteen samples are controlled hard poster/sign expansions rather than independently photographed real photos. This set is useful for strengthening failure conditions, but it should not be overstated as a statistical sample representing all real Korean environments. Third, local Canva/MiriCanvas templates were used for transcription and evaluation,

but the original images are not included in GitHub because of license and publication-scope constraints.

Future work should collect 50 to 200 real Korean photos in a balanced way across subway, poster, sign, and notice categories, and should annotate both ROI and reading order. A full ablation that removes each component one by one, bootstrap confidence intervals, and paired significance tests should also be added. The library could be made more automatic by adding vertical-writing detection, column detection, character-level crop/unwarp, and a layout classifier that works without vocabulary.

10. Conclusion

This study proposed a method that converts DeepSolo's ordered-points/boundary representation into Apple Vision OCR preprocessing and postprocessing strategies to improve reading-order errors in Korean vertical writing, multi-column notices, and dispersed typography images. The Core hard set, targeted synthetic set, and title-critical probe confirmed the effectiveness of pre-OCR proposals and ordered-point correction, while ordinary horizontal posters showed that the auto strategy avoids damage by preserving raw order. Therefore, DeepSolo is not merely a comparison baseline; it is the theoretical basis that reframes Apple Vision OCR outputs as points, boundaries, and ordered text units, and this interpretation is extended into the HangulGuidedOCR library.

References

- [1] M. Ye et al., DeepSolo: Let Transformer Decoder with Explicit Points Solo for Text Spotting, CVPR, 2023.
- [2] B. Shi, X. Bai, and C. Yao, An End-to-End Trainable Neural Network for Image-based Sequence Recognition and Its Application to Scene Text Recognition, IEEE TPAMI, 2017.
- [3] B. Shi et al., ASTER: An Attentional Scene Text Recognizer with Flexible Rectification, IEEE TPAMI, 2019.
- [4] Y. Baek et al., What Is Wrong With Scene Text Recognition Model Comparisons? Dataset and Model Analysis, ICCV, 2019.
- [5] M. Liao et al., TextBoxes: A Fast Text Detector with a Single Deep Neural Network, AAAI, 2017.
- [6] X. Zhou et al., EAST: An Efficient and Accurate Scene Text Detector, CVPR, 2017.
- [7] Y. Baek et al., Character Region Awareness for Text Detection, CVPR Workshops, 2019.
- [8] M. Liao et al., Real-time Scene Text Detection with Differentiable Binarization, AAAI, 2020.
- [9] Y. Liu et al., ABCNet: Real-time Scene Text Spotting with Adaptive Bezier-Curve Network, CVPR, 2020.
- [10] W. Zhang et al., TESTR: Text Spotting Transformers, CVPR, 2022.
- [11] Z. Wang, Y. Xu, L. Cui, J. Shang, and F. Wei, LayoutReader: Pre-training of Text and Layout for Reading Order Detection, EMNLP, 2021.
- [12] M. S. Kasem, M. Mahmoud, M. F. Senussi, M. Abdalla, and H. S. Kang, KORIE: A Multi-Task Benchmark for Detection, OCR, and Information Extraction on Korean Retail Receipts, Mathematics, 14(1):187, 2026.
- [13] G. Kim, J. Son, K. Lee, and J. Min, Character Decomposition to Resolve Class Imbalance Problem in Hangul OCR, arXiv:2208.06079, 2022.
- [14] Apple Developer Documentation, Vision Framework, VNRecognizeTextRequest, accessed June 8, 2026.
- [15] R. Smith, An Overview of the Tesseract OCR Engine, ICDAR, 2007.
- [16] JaidedAI, EasyOCR GitHub repository, first public release around 2020, accessed June 8, 2026.
- [17] NAVER Cloud Platform, CLOVA OCR product documentation, accessed June 8, 2026.
- [18] Facebook Research, Detectron2 documentation and GitHub repository, accessed June 8, 2026.
- [19] ViTAE-Transformer, DeepSolo official GitHub repository, accessed June 8, 2026.
- [20] wjdalswl, HangulGuidedOCR: Korean Reading-Order Correction Swift Package Based on Apple Vision OCR, GitHub repository, accessed June 8, 2026. <https://github.com/wjdalswl/HangulGuidedOCR>